Predicting Booking Cancellation at Hotel Haven

A Data Driven Approach to Proactive Revenue and Capacity Planning

10Alytics Python Machine Learning Capstone Project

Oghenevurie Lauretta

Business Background

Hotel Haven is a luxury hotel group operating across multiple locations in a competitive hospitality market.

The hotel experiences a high volume of **booking cancellations**, particularly for advance reservations, creating uncertainty in occupancy and revenue planning.

For a luxury brand, late cancellations have a disproportionate impact, limiting the ability to resell premium rooms and optimise staffing and service delivery.

There is a clear need to shift from **reactive cancellation handling** to **proactive, data-driven risk management**.

Project Objective & Key Business Questions

Project Objectives

- Understand the key booking and customer characteristics associated with cancellation behaviour.
- Evaluate whether historical booking data can reliably predict cancellation outcomes.
- Identify a robust modelling approach to support proactive cancellation risk management.
- Translate analytical insights into practical, operational recommendations for Hotel Haven.

Key Business Questions

1. What booking and customer characteristics influence cancellation behaviour?
2. Can booking outcomes be predicted reliably using historical data?
3. Which factors most strongly drive cancellation risk?
4. How can these insights be used to reduce cancellations and improve planning?

Data Overview

The analysis is based on historical booking data from Hotel Haven, covering customer information, booking characteristics, pricing, stay patterns, and booking outcomes.

The dataset captures the full booking lifecycle, from reservation timing through to cancellation or completed stay.

Initial review confirmed strong data completeness and consistent structure, enabling focus on feature validation and modelling rather than missing data remediation.#

The data provides sufficient depth and variety to support both behavioural analysis and predictive modelling of cancellations.

Data Cleaning & Preparation

- Validated and encoded the target variable to ensure a consistent and reliable definition of booking cancellation.
- Conducted feature-level review to assess logical consistency, frequency distribution, and modelling suitability.
- Addressed extreme values and low-frequency categories through capping and consolidation to improve model stability.
- Simplified high-cardinality and skewed features into interpretable indicators while preserving behavioural signal.
- Converted date fields into structured formats to support temporal analysis.

Feature Engineering and Preprocessing

- ❑ Separated the validated dataset into feature variables and the target outcome for modelling.
- ❑ Categorised features into numerical, categorical, binary, and time-derived variables.
- ❑ Applied appropriate encoding techniques to categorical features to support model compatibility.
- ❑ Scaled numerical variables where required to ensure fair model comparison.
- ❑ Split the dataset into training and test sets using an 80/20 split with a fixed random state to ensure reproducibility.

Predictive Modelling Strategy

The problem was framed as a **binary classification task**, predicting whether a booking would be cancelled.

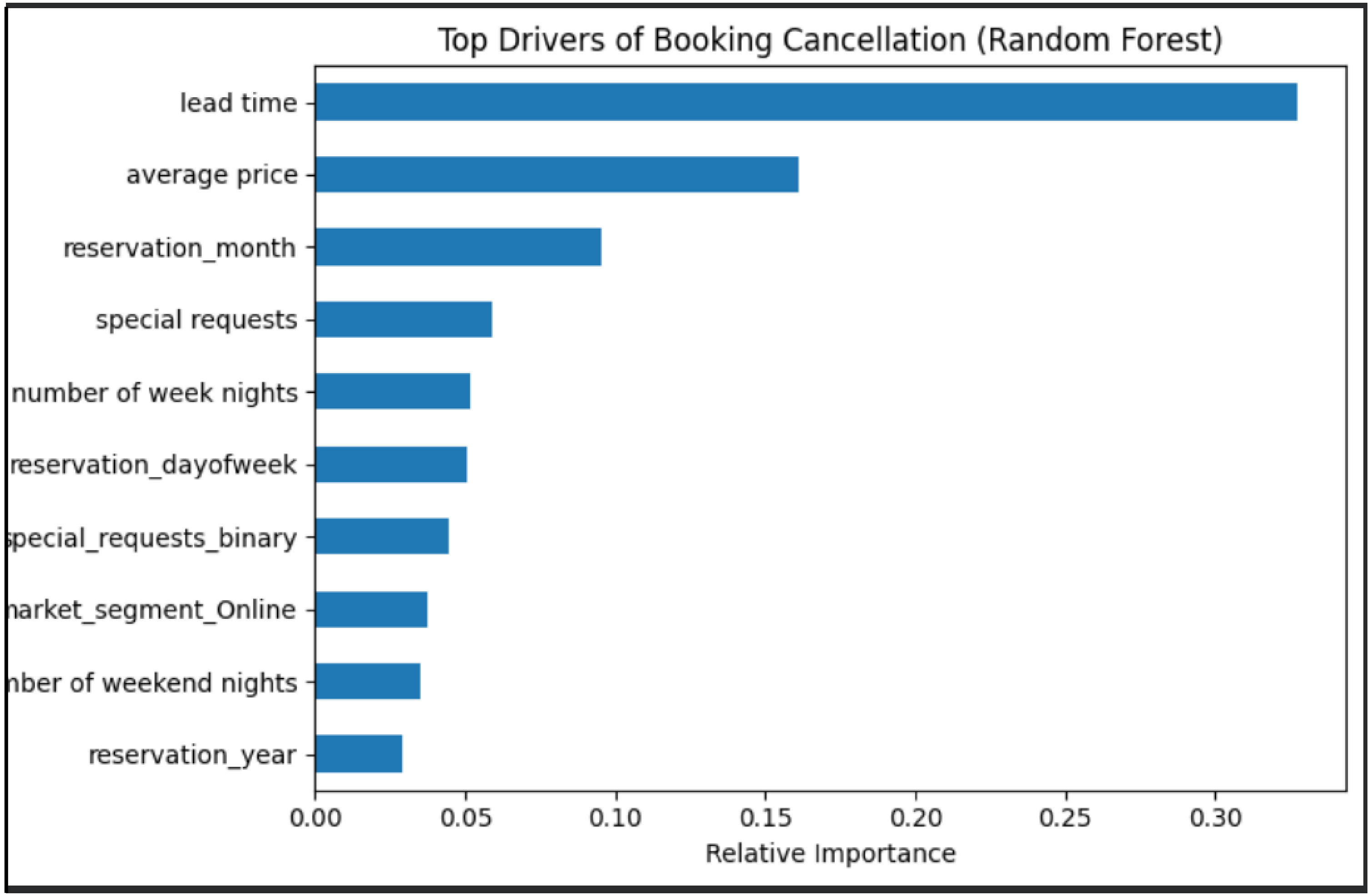
Multiple supervised learning models were evaluated to balance interpretability and predictive performance.

A progressive modelling approach was adopted:

- **Logistic Regression** as a transparent baseline
- **Decision Tree** to capture non-linear relationships
- **Random Forest** and **Gradient Boosting** as ensemble methods to improve robustness

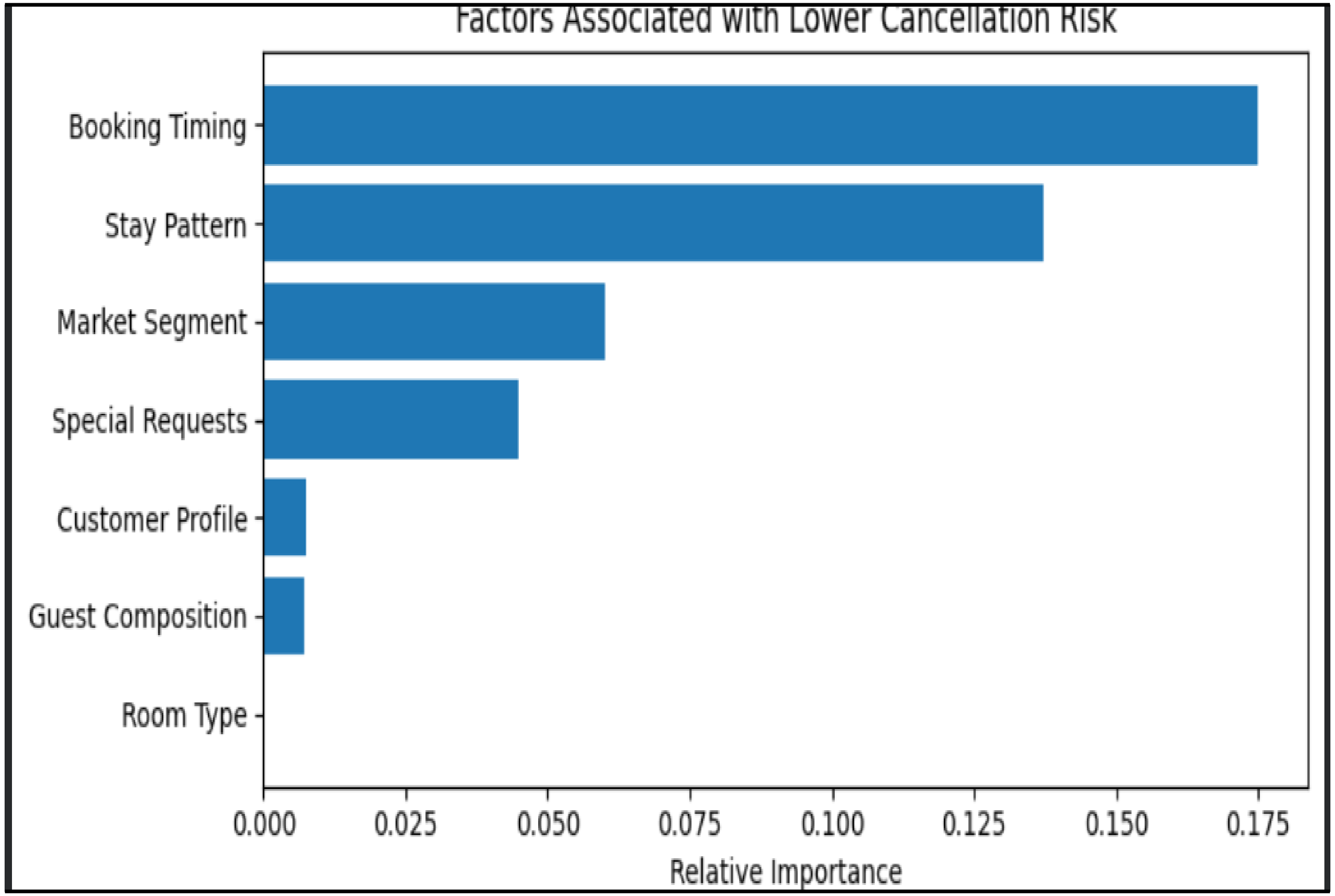
Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices, with particular emphasis on **recall for cancelled bookings** due to its business impact.

Business Question 1: What Influences Cancellation Behaviour?



- **Lead time is the strongest risk driver** – bookings made far in advance are less stable and more likely to cancel.
- **Higher average price increases cancellation risk**, suggesting stronger price sensitivity at premium rates.
- **Booking timing (seasonality/calendar effects)** contributes to risk variation across the year.
- Lower engagement signals (e.g., **fewer special requests**) are associated with reduced commitment and higher cancellation probability.
-

Business Question 1: What Influences Cancellation Behaviour? (Stability Signals)



- Not all bookings carry equal risk – the model also identifies **features linked to higher booking stability**.
- **Booking timing** (shorter lead times) is the strongest signal of commitment, indicating guests are less likely to cancel.
- **Stay patterns** associated with structured or shorter stays show lower cancellation likelihood.
- **Market segment and special requests** reflect guest intent and engagement, reducing cancellation risk.

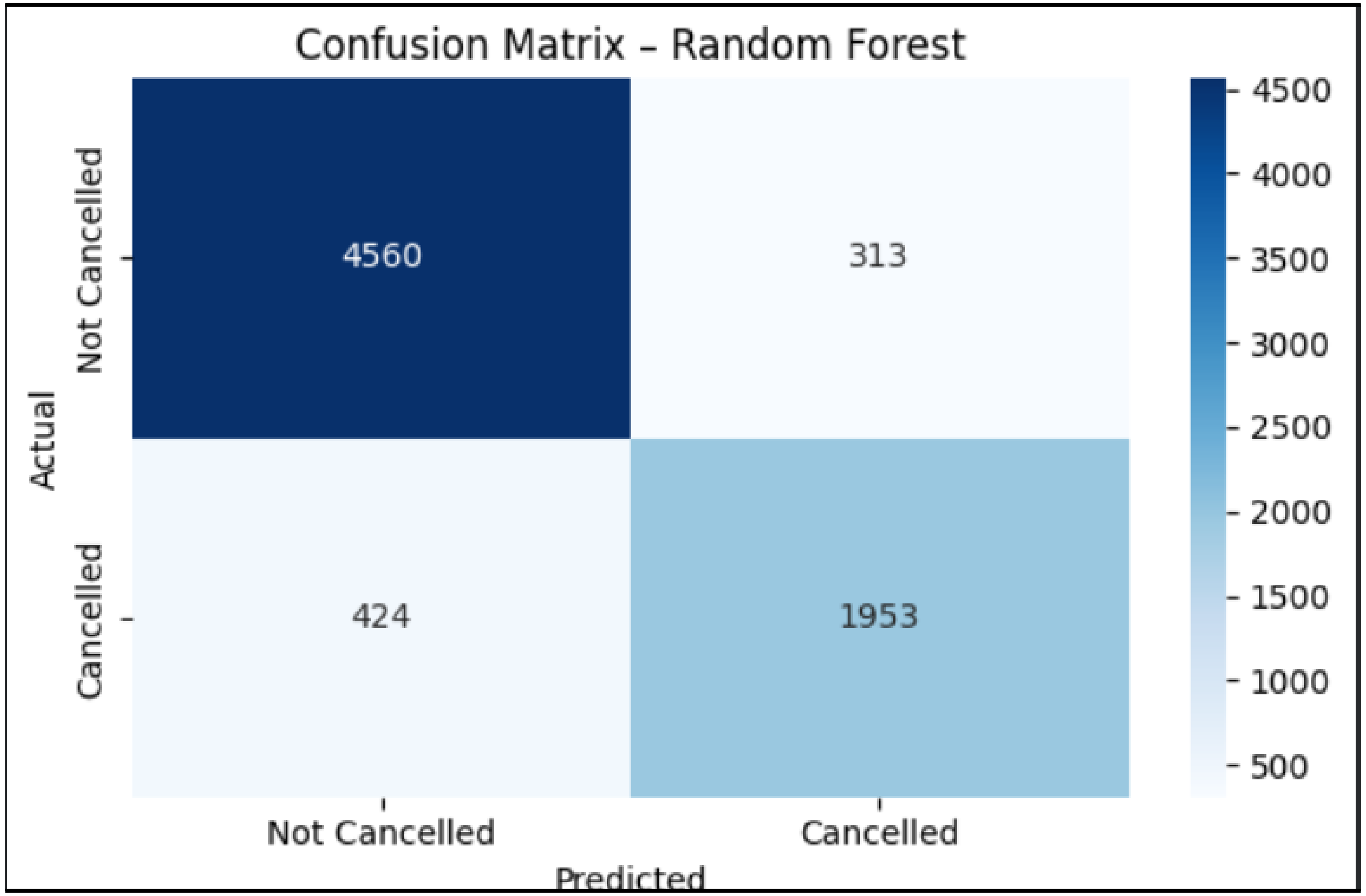
Business Question 2: Can Booking Outcomes be Predicted Reliably using Historical Data?

Model Performance Comparison

Model	Accuracy	Precision (Cancelled)	Recall (Cancelled)	F1-Score (Cancelled)
Logistic Regression	0.81	0.75	0.64	0.69
Decision Tree	0.83	0.77	0.71	0.74
Random Forest	0.90	0.86	0.82	0.84
Gradient Boosting	0.85	0.81	0.70	0.75

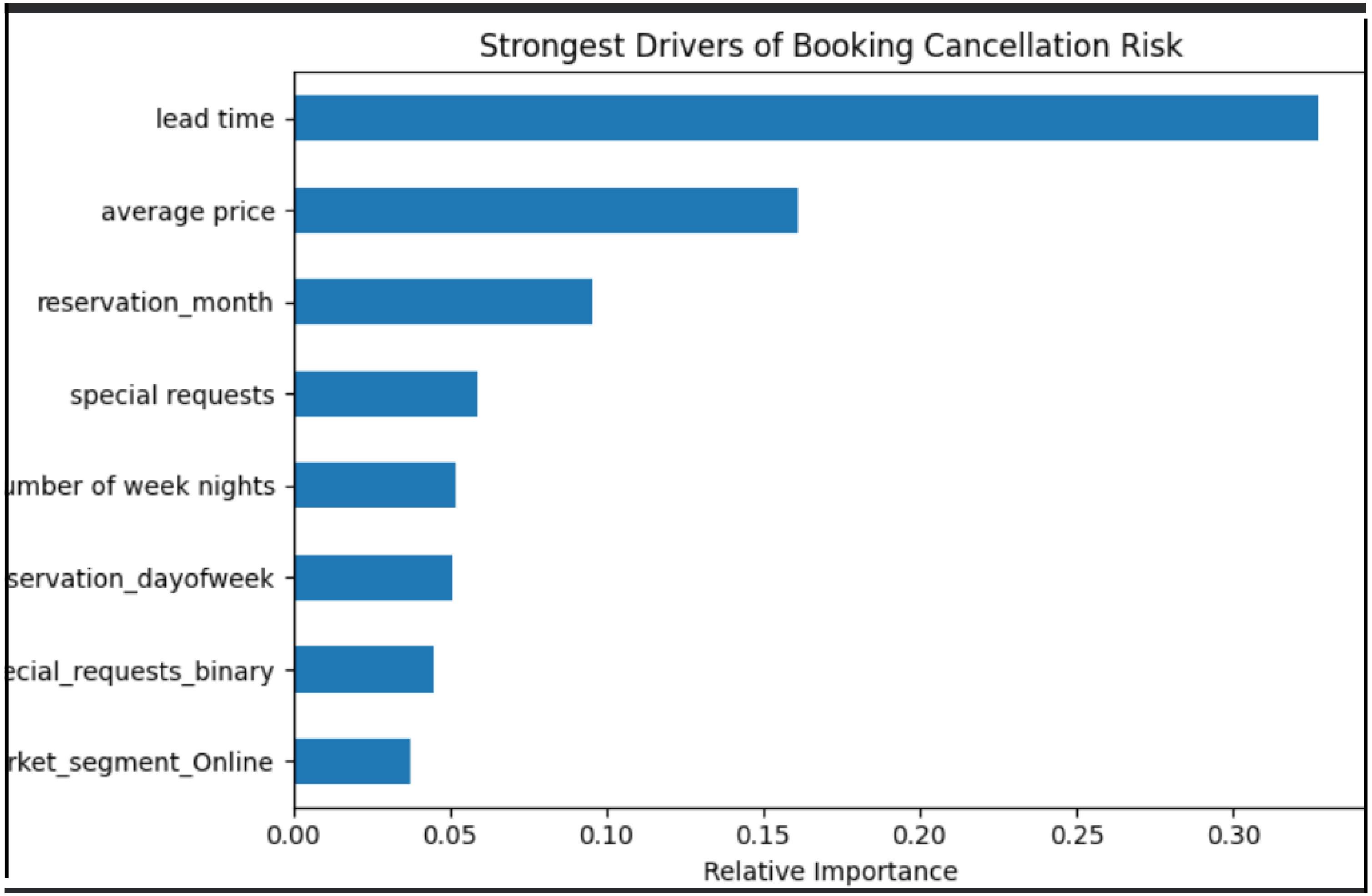
Business Question 2 - Validation & Confidence

Model Validation: Confusion Matrix



- **Correctly identified 1,953 cancelled bookings**, enabling early intervention
- **Missed 424 cancellations**, representing the remaining risk exposure
- **Strong protection against false alarms**, with 4,560 non-cancelled bookings correctly retained
- **Balanced performance**, minimising both revenue loss and unnecessary operational action

Business Question 3: Which Factors Most Strongly Drive Cancellation Risk?



- ❑ **Lead time** is the dominant driver of cancellations
  → bookings made far in advance carry the highest risk
- ❑ **Average price** is the second most influential factor
  → higher-priced bookings are more likely to cancel
- ❑ **Booking timing (reservation month)** reflects seasonal risk patterns
- ❑ **Special requests and stay structure** signal booking commitment
- ❑ **Market segment (online bookings)** shows elevated cancellation behaviour

Business Question 4: How Can These Insights Reduce Cancellations and Improve Planning?

Early risk flagging for long lead-time bookings
→ apply monitoring or flexible confirmation strategies.

Targeted interventions for high-risk segments
→ pricing-sensitive, online, and high-value bookings.

Use special requests as commitment signals
→ prioritise low-risk bookings in capacity planning.

Improve staffing and inventory forecasting
→ align demand planning with predicted cancellations

Enable proactive revenue protection strategies
→ overbooking controls, deposits, or targeted incentives

Conclusion

- ❑ Booking cancellations can be **predicted reliably** using historical booking data.
- ❑ **Ensemble models outperform baseline approaches**, with Random Forest delivering the strongest overall performance.
- ❑ **Lead time, pricing, booking timing, and customer behaviour** are the primary drivers of cancellation risk.
- ❑ Predictive insights enable **proactive intervention**, improved planning and revenue protection.
- ❑ The analysis provides a **scalable foundation** for data-driven cancellation management at Hotel Haven.

Limitations

Analysis is based on **historical booking data**, which may not capture future behavioural shifts.

External factors such as **economic conditions, competitor pricing, or weather events** are not included.

Model performance depends on **data quality and feature availability** at the time of booking.

Cancellation behaviour may evolve with **policy or customer incentive changes.**

Interpretability is reduced for **ensemble models** compared to simpler baseline models.

Future Work

1

Extend the model to **real-time booking risk scoring** at the point of reservation.

2

Integrate **external signals** (events, seasonality indicators, demand trends) to enhance prediction accuracy.

3

Explore **cost-sensitive optimisation** to explicitly balance missed cancellations against false alerts.

4

Apply **model explainability techniques** (e.g. SHAP values) to support operational trust and decision-making.

5

Embed the model within **revenue management or CRM systems** to enable automated, proactive interventions.

Thank you for your time.

Any Questions?